Thermal regulation loop Specification

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Revision	A
Status	APPROVED
Classification	INTERNAL
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Approved by	Selene Abaroth
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1. Purpose

This specification defines the performance, interface and verification requirements for the thermal regulation loop (LS-TRG) of the Life Support & Atmospherics on Starhaven Orbital Habitat & Shipyard.

2. Scope

Fabrication and installation are performed under work package WP-SHV-LS-05 (WBS 1.8.5), owned by Nerys Chalke.

The prime supplier for flight hardware is Fondor Foundry Collective. Responsible engineer: Tyce Easton.

3. Applicable documents

Document	Title
SHV-SYS-SPEC-0001	Starhaven Master System Specification
SHV-LS-DWG-0204	Thermal regulation loop general arrangement drawing
SHV-LS-BOM-0304	Thermal regulation loop bill of materials
SHV-QA-PLN-0004	Starhaven Verification and Test Plan
SHV-QA-STD-0002	Station Workmanship Standard
SHV-ICD-PD-LS	Interface Control Document: Power Distribution / Life Support & Atmospherics
SHV-ICD-LS-HA	Interface Control Document: Life Support & Atmospherics / Habitation Blocks
SHV-ICD-LS-WM	Interface Control Document: Life Support & Atmospherics / Waste Management & Recycling
SHV-ICD-LS-AG	Interface Control Document: Life Support & Atmospherics / Hydroponics & Food Production

4. Description

Life support keeps the atmosphere, water, gravity and temperature within crew limits in every inhabited zone.

The thermal regulation loop is delivered as a set of line-replaceable assemblies, integrated on station by Construction Operations and verified by Quality Assurance & Test.

5. Performance requirements

ID	Requirement	Value	Tolerance	Verification
LS-TRG-R01	The thermal regulation loop shall provide a setpoint of 21.8 °C.	21.8 °C	±0.5%	Analysis

5.1 General requirements

ID	Requirement	Verification
LS-TRG-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.	Inspection
LS-TRG-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.	Analysis
LS-TRG-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.	Inspection
LS-TRG-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.	Analysis
LS-TRG-G05	Documentation: Acceptance data packages shall be delivered with every unit.	Inspection

6. Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

7. Verification

Each requirement is verified by the method stated in section 5, following SHV-QA-PLN-0004. Acceptance test reports are filed under the LS test-report series.

8. Revision history

Rev	Date	Description	Author
A	2023-09-12	Initial release.	Tyce Easton